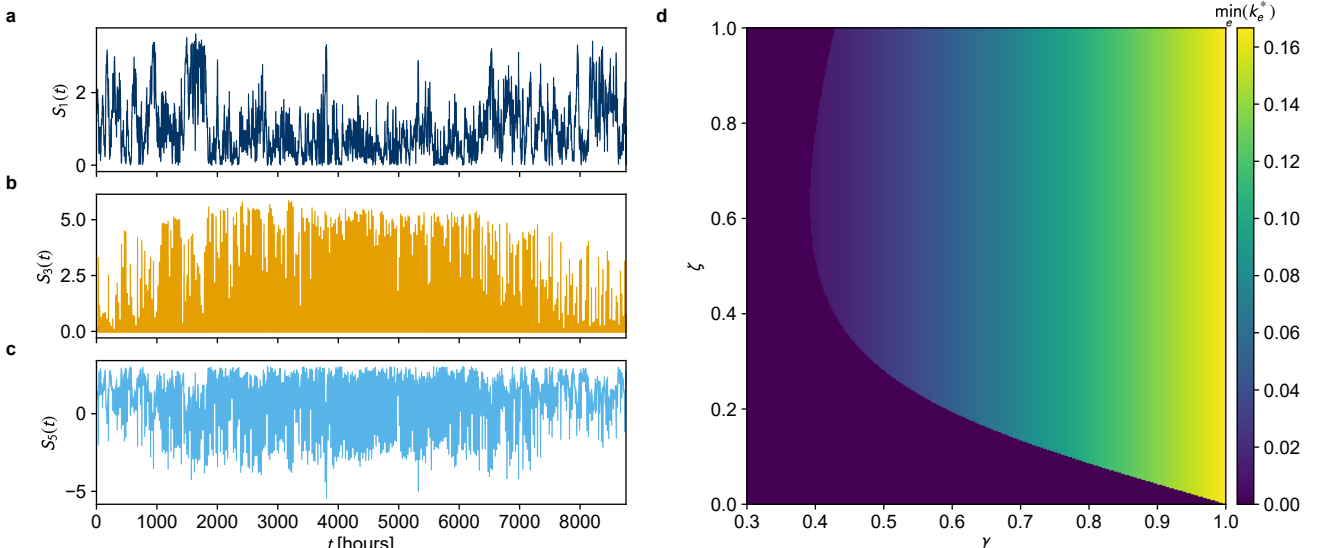

a
b
c
d
$S_1(t)$
$S_2(t)$
$S_3(t)$
2
0
5.0
2.5
0.0
−5
0
1000
2000
3000
4000
5000
6000
7000
8000
t [hours]
$\min_e(k_e^*)$
1.0
0.8
0.6
0.4
0.2
0.0
ζ
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
γ
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00